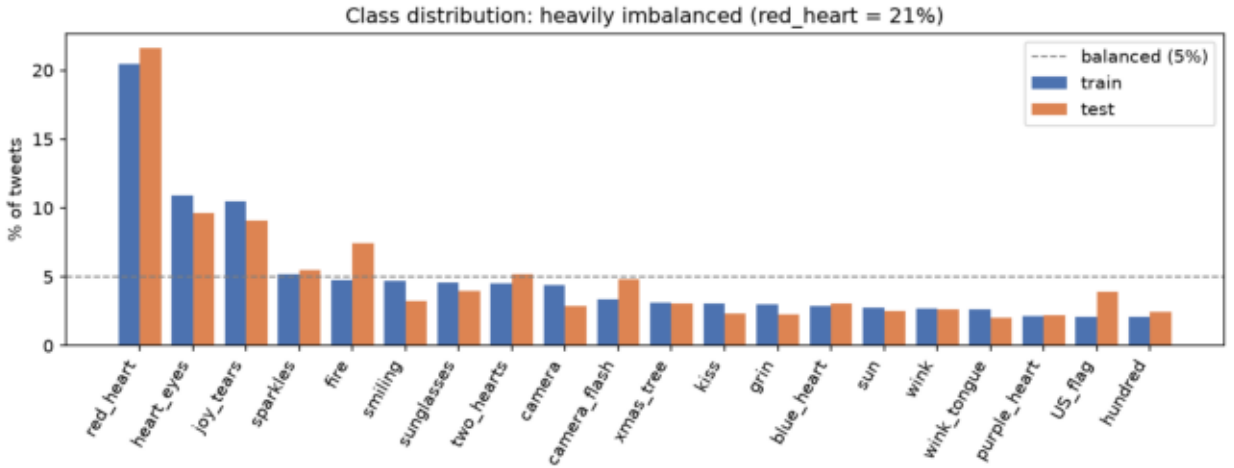
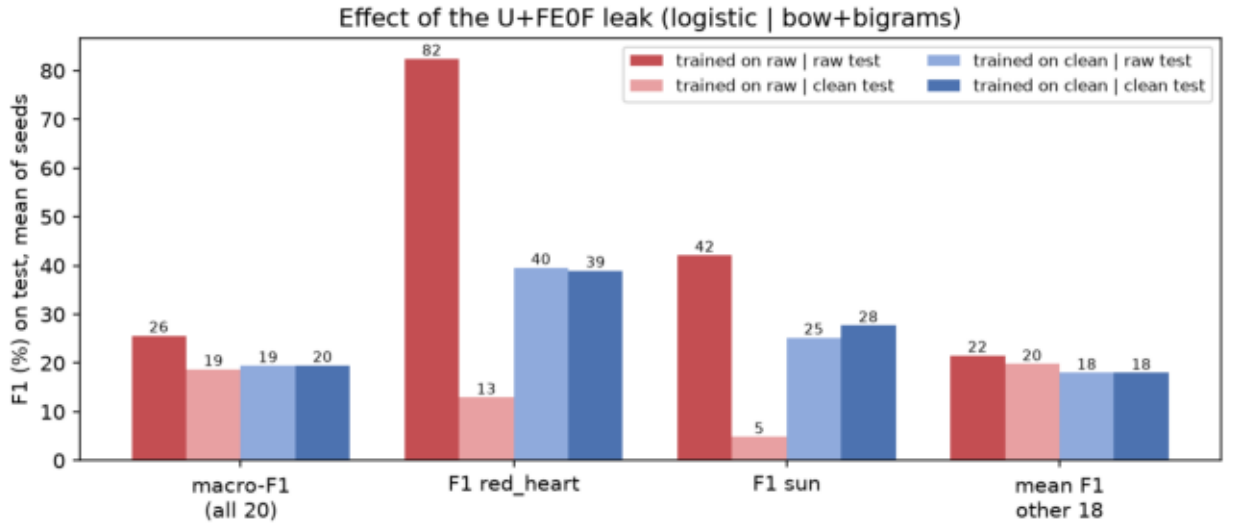
1 Dataset: TweetEval emoji (20 classes)

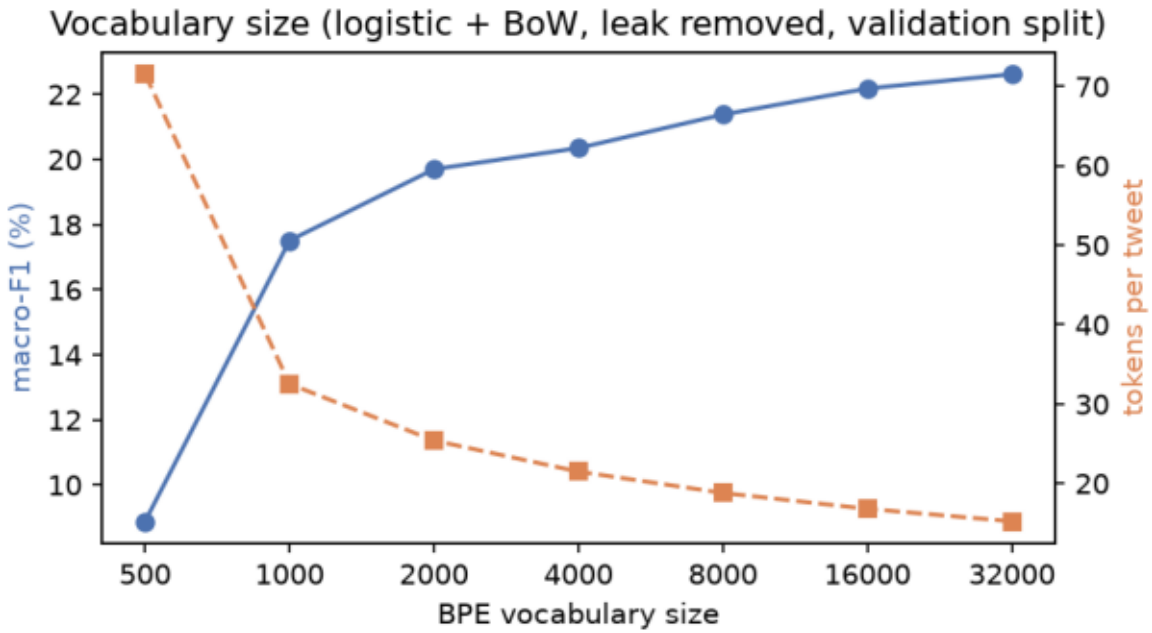
- Task: guess which of 20 emojis a tweet contained (emoji deleted from the text)
  - 45k train (CV) · 5k validation (choices) · 50k test (final); median 12 words
- Very imbalanced: red_heart = 21% of tweets, hundred / US_flag ≈ 2%
  - 'always red_heart': 21.6% accuracy, 1.8% macro-F1 → we use macro-F1
- Label leak: deleting ❤️ ☀️left an invisible character (U+FE0F)
  - in 20% of train tweets (86% red_heart); in train + test, not validation
  - macro-F1 25.6 with leak vs 19.5 without (about 6 fake points)
  - trained with leak, tested without: red_heart F1 13 → shortcut learning
- Decision: remove U+FE0F from all splits





2 Tokenizer: BPE trained on our tweets

- New tokenizer? Yes: tweets (hashtags, slang, @user) differ from the tutorial's news
- Data: the 45k training tweets, text only (no labels, never test); ~530k words is enough
- Hyperparameters: Metaspace pre-tokenizer, min_frequency = 2 (tutorial); vocab size tested:
  - each size: BPE → bag-of-words → logistic regression → macro-F1 on validation
  - diminishing returns: 16k = 22.2, 32k = 22.6; at 32k, 13,234 tokens occur in < 5 tweets
  - 500 = character level: tweets contain ~557 distinct characters, so no merges happen
- Decision: vocabulary = 16,000 (within 0.5 pt of 32k, half the dimensions)
  - '#blessed @ San Diego' → _#blessed _@ _San _Diego



3 Features: three representation spaces

- Every tweet becomes a vector with one position per feature; stored sparsely (≈ 17 non-zeros out of 16,000)
- Token indicator (tutorial): 1 if the token occurs, else 0
- Bag-of-words (tutorial): how many times each token occurs
  - repetition can carry emotion: 'so so happy', 'love love love'
- Ours, BoW + bigrams: also counts pairs of neighbouring tokens
  - keeps local word order: 'not good' ≠ 'good'; '_San+_Diego', '_best+_friend' become single features
  - only the 20,000 most frequent pairs (seen ≥ 3 times in training) get a dimension; all V^2 = 256M pairs would be far too many
  - still misses long-range order ('not … at all good') and gets bigger and sparser
- All three use the same 16k BPE tokenizer, so differences come from the representation only
- → compared with 3 architectures in cross-validation (slides 4–6)

```
tweet: "so so happy with my best friend"

token indicator  (16,000 dims, 6 non-zeros)
  _so=1 _happy=1 _with=1 _my=1 _best=1 _friend=1

bag-of-words     (16,000 dims, 6 non-zeros)
  _so=2 _happy=1 _with=1 _my=1 _best=1 _friend=1

BoW + bigrams    (36,000 dims, 12 non-zeros)
  _so=2 _happy=1 _with=1 _my=1 _best=1 _friend=1
  _so+_so=1 _so+_happy=1 _happy+_with=1
  _with+_my=1 _my+_best=1 _best+_friend=1
```